

Hydrozoan Artifact: Lean Formalization

This note accompanies the paper *Hydrozoan: Latency-Adaptive DAG Consensus under Mixed Byzantine and Crash Faults*. Sections, lemmas, theorems, algorithms, tables and figures numbered without the prefix “L” refer to the paper.

L. Lean Formalization

The lemmas and theorems of Appendices A and B and the liveness of both protocols are machine-checked in Lean 4 on top of Mathlib.¹ Each protocol is one module of the development: `LeanDag/Hydrozoan` for **Hydrozoan**, and `LeanDag/OptimalHydrozoan` for **Optimal-Hydrozoan**, which imports the first read-only. Both modules follow the same partition. `Model/` holds definitions and no theorem. Each result is a proposition `Statement` in its own `Statement.lean`, which holds no proof, and is proved by the theorem holds in the sibling `Proof.lean` with lemmas from `Helpers/`. A script of the repository (`scripts/check-arc-holes.py`) enforces the partition and rejects `sorry`, `admit`, `axiom` and `native_decide` anywhere in either module.

A reviewer audits `Model/` and the statement files, about a thousand lines for each of **Hydrozoan** and **Optimal-Hydrozoan**, together with the shared definitions they import from `LeanDag/Common` (the block record, the leader schedule, and the anchored decision relation). The proofs and helpers, a few thousand lines each, are AI-generated and need no reading: the kernel checks them against the audited statements. Every top-level theorem depends on no axioms beyond Lean’s three standard ones (`propext`, `Quot.sound` and `Classical.choice`), and the build enforces it: a `#print axioms` guard per result fails compilation if the list changes. A further few thousand lines of witnesses evaluate every definition by `decide` on concrete DAGs before any theorem uses it, guarding against vacuous statements. They include a DAG on which a slot fast-commits while no certificate for it exists anywhere, so the weak rung of the graded indirect rule is necessary; a crash-only committee at the bound ($n = 5$, $c = 2$) on which per-slot leader rotation never yields three consecutive correct leaders; and, for **Optimal-Hydrozoan**, the attack of FinWhale [1] on the direct skip of a slot whose Byzantine leader proposed. In Table L1, names are relative to `LeanDag.Hydrozoan` (top) and `LeanDag.OptimalHydrozoan` (bottom), except the shared `AnchoredRule` in `LeanDag`; capitalized names are audited `Statement.lean` results or their conjuncts, lowercase names are proved lemmas.

1. <https://anonymous.4open.science/r/lean-dag/226b9155>

L.1. The model

The fault model is a class `Faults` carrying the bounds f , c and Hydrangea’s $k = 2p - c$, the actual Byzantine and crashed sets, disjoint and within their bounds, and the committee bound $n \geq 3f + 2c + k + 1$. The five thresholds of Table 1 are definitions over it, with $p = \lfloor (c + k)/2 \rfloor$ derived rather than supplied (so $n \geq 3f + c + 2p + 1$ holds by construction), so an inadmissible p cannot be expressed. **Optimal-Hydrozoan** extends the class with $f + c \geq 1$ and defines $p + 1$, q_{plain} and q_{equiv} (`tPlain` and `tEquiv` in `Lean`) on top of it.

A *block universe* holds every block accepted by the DAG-building layer of any validator, Byzantine ones included. Every block references at least q blocks of distinct authors, all from the previous round, and no non-Byzantine validator, correct or crashed, authors two blocks in one round. A *view* is one validator’s local DAG: a subset of the universe closed under references. Validators thus disagree only on which blocks they hold, and Byzantine withholding is a block of the universe missing from a view. For **Optimal-Hydrozoan**, the universe adds the validity rule of Section 6 as a clause on every block.

The rules of Algorithms 1 and 2 are predicates on a view that count distinct authors against the thresholds. The decision relation `Decided( $V, s, v$ )`, for a view V , a slot s and a verdict v (a block or skip), is inductive with four constructors: fast or direct commit, direct skip, indirect commit at the first nonempty rung from the nearest eligible committed anchor, and indirect skip when every rung is empty. The relation places no order between constructors: any justified verdict is derivable, and `SlotAgreement` proves that no two derivations disagree. The weak rung breaks ties by a linear order on block identifiers, standing for the minimum digest.

L.2. Statements

Safety is stated over every fault configuration, leader schedule, block universe and pair of views the model admits, and assumes nothing about the network. `SlotAgreement` quantifies over any two verdicts on one slot, so it covers the fast and direct commits, the direct skip, the certificate and weak rungs and the indirect skip at once. `PrefixAgreement` lifts it to the committed-leader sequences read off in slot order, and `Delivery` to the delivered sequences.

Liveness avoids clocks by treating synchrony structurally. A set T of correct validators with $|T| \geq q$ populates a

TABLE L1. PAPER STATEMENTS AND THEIR LEAN COUNTERPARTS.

Paper statement	Lean declaration	Content
<i>Hydrozoan (Appendix A)</i>		
Lemma 1	ThresholdArithmetic; qWeak_le_qCert, qSlow_le_q, n_add_qWeak_add_f_le_qFast_add_q certifiedIn_of_slowCommit	Every relation for every committee $n \geq 3f + c + 2p + 1$, with no cap on p ; $q_{weak} \leq q_{cert}$, $q_{direct} \leq q$ and the last row in its form with f are lemmas.
Lemma 2		A direct commit's certificate is in the causal history of every block above the decision round.
Lemma 3	weakLinked_of_fastCommit	q_{weak} votes of a fast commit are in the causal history of every block from the decision round on.
Lemma 4	DirectSafety.CertUniqueness	Two certified candidates of one slot are one block.
Lemma 5	DirectSafety.CommitSkipExclusion; SlotAgreement.hydrozoanLaws	A fast or direct commit in one view excludes a direct skip in any view; a direct skip empties both commit rungs.
Lemma 6	DirectSafety (fast/fast, direct/direct, fast/direct); SlotAgreement.hydrozoanLaws	Two fast or direct commits name one block; every eligible anchor links such a commit and links no rival first.
Lemma 7	SlotAgreement.hydrozoanLaws	One anchor, one rung and the tie-break select one block.
Lemma 8	SlotAgreement	Any two verdicts on one slot, from any two views and by any of the six routes, are equal.
Lemma 9	PrefixAgreement	Committed-leader sequences agree at equal horizons and are prefixes at different ones.
Lemma 10, Theorem 11	Delivery.DeliveredPrefixConsistency; PrefixAgreement; AnchoredRule.slot_eq_of_decided_commit	Delivered sequences are prefixes of one another across views and horizons; a committed block belongs to one slot.
Theorem 12	Delivery.Integrity; Delivery.Faithful	No block is delivered twice, and none is lost.
Lemma 13	DirectLiveness	A synchronized wave with a correct leader is directly committed.
Theorem 15	EventualDecision; Delivery.Faithful	A committed run decides every slot below it, and every block of the ledger is delivered; the step placing an honest block in a later committed leader's history is argued in Appendix A.
Lemma 14, Theorem 16	IndirectLiveness, EventualDecision; Delivery.DeliveredPrefixConsistency	A committed run decides every slot below it; fair schedules place such runs past every slot; delivered sequences are prefixes of one another.
Fair schedule	Grounding	Wave-aligned rotation is fair with no premise; the liveness hypotheses are satisfiable at every horizon.
<i>Optimal-Hydrozoan (Appendix B)</i>		
Lemma 17	ThresholdArithmetic	Every row, including the one guarded by $f \geq 1$, under $f + c \geq 1$.
Lemma 18	leader_byzantine_of_witnesses; Grounding	A witnessed equivocation names a Byzantine leader; a synchronized universe authored by a quorum satisfies the validity rule at every horizon.
Lemma 19	isFastEvidence_exclusive	A block is fast evidence for at most one candidate.
Lemma 20	isFastEvidence_of_fastCommitOpt	Under the validity rule, every decision-round block is fast evidence for a fast-committed block.
Lemma 21	evidenceLinked_of_fastCommitOpt	Every block above the decision round links an evidence quorum for it.
Lemma 22	evidenceLinked_unique	One anchor links an evidence quorum for at most one candidate.
Lemma 23	DirectSafety.CommitSkipExclusion; SlotAgreement.optimalLaws	As for Hydrozoan, with the skip of Algorithm 5.
Lemmas 24 and 25	DirectSafety; SlotAgreement.optimalLaws; SlotAgreement	As for Hydrozoan; the evidence rung needs no tie-break.
Prefix, delivery and liveness	PrefixAgreement, Delivery, DirectLiveness, IndirectLiveness, EventualDecision, Grounding	As for Hydrozoan; in addition, a slot with no candidate is directly skipped by any quorum of correct validators.

round when each of them authors a block in it, and is *synchronized* from round R when every block of T references every block of T in the round below. `DirectLiveness` concludes that a populated, synchronized wave whose leader is in T is directly committed in any view that reaches its decision round. The guarantee goes through the direct path because only q validators are certain to vote, and $q < q_{fast}$ in general. The fast commit and the direct skip of `Hydrozoan` appear as separate theorems that hold only when the actual number of faults is at most p , outside the liveness statement. For `Optimal-Hydrozoan` the direct skip of a slot with no candidate is part of the liveness statement.

L.3. Trusted base

Beyond the kernel, a reader trusts the audited definitions and the following modeling choices. A vote is a direct reference, which coincides with `ISVOTE` at wave length 3 by an argument in prose, not in Lean; the filtering of malformed

blocks by the DAG-building layer is assumed. The derivation of synchrony and population from message delivery, `GST` and timeouts is out of scope, so the liveness theorems are exactly as strong as these two hypotheses; `Grounding` shows the hypotheses satisfiable at every horizon, not that a network realizes them.

References

- [1] R. Ladelsky and R. Friedman, “FinWhale: An optimally resilient two-round terminating DAG protocol,” in *DISC*, 2026.